

MAT 100 Precalculus

Homework 3

Basic Information

This assignment is due on Gradescope by **1:30 PM on Monday, September 21**.

Make sure you understand MHC [honor code](#) and have carefully read and understood the additional information on the [class syllabus](#). I am happy to discuss any questions or concerns you have!

A component of this class is helping you understand *why* the math you use works the way it does. To that end, make sure you show all your work as you will be graded on the *process* you use, not just your final answer. And if a question asks you to explain *why* something is true, be sure to answer that part of the question in complete sentences.

Homework will be graded for correctness, and I will point out where mistakes or errors are made. Each homework will be graded *Excellent*, *Satisfactory*, or *Unsatisfactory*. The [course rubric](#) explains more about the distinctions of these categories.

Turn-In Problems

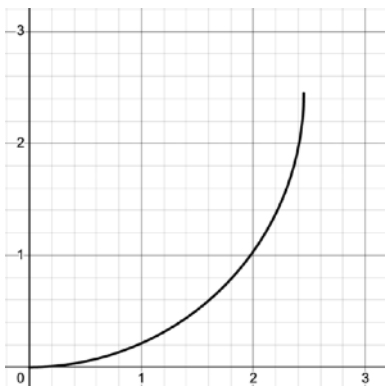
1.1: 36

1.2: 2, 10, 16

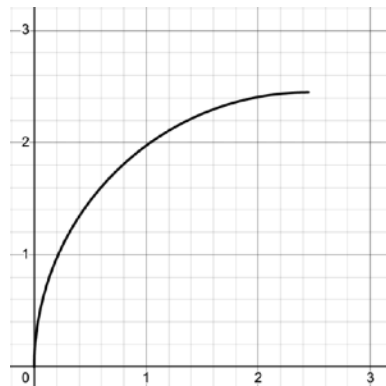
1.3: 24, 26, 34, 38 (you can use Desmos for the last problem!)

1.4: 2, 22

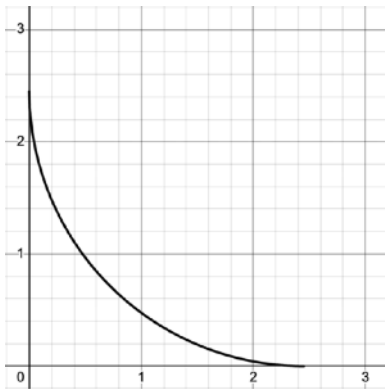
11. A function can increase quickly or slowly, and similarly for a decreasing function. The idea of how fast the graph is changing is called *concavity*. Increasing and decreasing can each be paired with concave up or concave down. That gives us 4 different possible “looks” for a graph (See A, B, C, and D below— 2 are on the next page). For each of these 4 pictures, come up with a real world example (like from our examples in class Thursday which you can find in Canvas under “Files/Class Slides”) that have a graph of these general shapes.



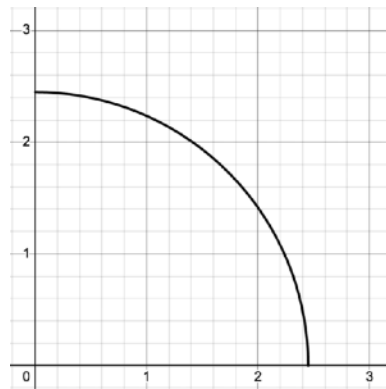
A. concave up and increasing



B. concave down and increasing



C. concave up and decreasing



D. concave down and decreasing

Additional Problems (to do on your own, not to turn in)

1.1: 35

1.2: 7, 15

1.3: 23, 33, 37 (you can use Desmos for the last problem!)

1.4: 1, 21